Midterm Review

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Roadmap for today

A few general review bullet points and practice problems in advance of the midterm

Bias/Variance:

- Metrics:
  - In regression problems, we have the expected MSE, the training MSE and the test MSE.
  - In classification problems, we have the expected error rate, the training error rate and the test error rate.
- Model selection:
  - The best model yields the smallest expected (test) MSE (error rate).
  - Among models that have similar expected MSE (error rate), we always prefer the more parsimonious one.
- Bias and variance trade-off:
  - A more complex / flexible $\hat{f}$ has smaller bias but larger variance

Bias/Variance

Practice Question (5 min: Suppose we have a data generating process $Y = f(X) + \epsilon$, where $\mathbb{E}[\epsilon] = 0$, $\mathbb{E}[\epsilon^2] = \sigma^2$. For a fixed test point x_t, let $\hat{f}(x_t)$ be an estimate of $f(x_t)$ constructed from a training data set D. Show

$$\mathbb{E}_{D,\epsilon}\left[(Y - \hat{f}(x_t))^2\right] = \text{Bias}[\hat{f}(x_t)]^2 + \text{Var}(\hat{f}(x_t)) + \sigma^2$$

Model Selection:

Two approaches to consider for model selection:

1. Estimate the expected MSE by "holding out" a portion of your training data for validation:
   - Validation set approach
   - Cross-validation set approach
2. Make an adjustment to the training error to penalize more complicated models:
   - AIC and BIC
   - Adjusted R^2
   - Mallow's C_p

Model Selection

Practice Question (5 min): Let MSE_i be the error on the ith fold in cross-validation. The CV estimate is $\mathrm{MSE}_{\mathrm{CV}} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i$.

1. Why is the bias of the LOOCV estimate generally lower than that of 5-fold CV?
2. In terms of the variance of the estimate, $\mathrm{Var}(\mathrm{MSE}_{\mathrm{CV}})$, why might LOOCV perform worse than 10-fold CV? (hint: Consider the correlation ρ between MSE_i and MSE_j).

Shrinkage Regression:

We can fit a model containing all p predictors using a technique that constrains or regularizes the coefficient estimates by shrinking the coefficient estimates towards zero:

- Linear Regression uses an ordinary least squares objective: $\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i^\top \beta)^2$
- Ridge Regression uses an ℓ_2 penalty term as well: $\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$
- Lasso Regression uses an ℓ_1 penalty term as well: $\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i^\top \beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$

Neither ridge regression nor the lasso universally dominates the other.

Shrinkage Regression

Practice Question (5 min): How does the bias-variance tradeoff play out in terms of comparing a Ridge and a Lasso estimator? Which method is expected to have lower variance in the presence of many irrelevant predictors, and why?

Nonlinear Regression:

To move beyond linearity, we introduced a few methods:

- Univariate ($p = 1$):
  - Polynomial Regression
  - Step Functions
  - Splines
- Multivariate ($p > 1$):
  - (Weighted) K-Nearest Neighbors
  - Local Regression
  - Generalized Additive Models

Nonlinear Regression:

Practice Question (5 min): Consider a cubic spline $S(x)$ on an interval $[a, b]$ with K interior knots $\xi_1, \ldots, \xi_K$.

1. Explain why a piecewise cubic polynomial with no continuity constraints would have $4(K + 1)$ parameters
2. Suppose we impose continuity constraints (i.e., continuity of the function, first derivative, and second derivative). How many degrees of freedom does the model have? (hint: think about the number of constraints we introduced)

Gradient Descent

Suppose we have an optimization problem of the form

$$\hat{w} = \arg\min_{w} J(w)$$

When we do not have a direct solution to $\frac{\partial J}{\partial w} = 0$, we can use gradient-descent to iteratively find an optimum. At iteration $k+1$, for each $j \in \{1, \ldots, p\}$:

$$w_j^{(k+1)} \leftarrow w_j^{(k)} - \alpha \left. \frac{\partial J}{\partial w_j} \right|_{w=w^{(k)}}$$

Gradient Descent

Practice Question (5 min): Let $g_i = \nabla_w J_i(w)$ be the gradient for observation i. Assume g_i are i.i.d. with variance $\text{Var}(g_i) = \sigma^2 I$.

1. Show that the variance of the mini-batch update direction, $\bar{g}_m = \frac{1}{m} \sum_{i \in B} g_i$ (where $B \subset \{1, \ldots, n\}$ is a mini-batch of size m), is $\frac{\sigma^2}{m} I$.
2. Explain the "Computational vs. Statistical" trade-off: Why don't we just use $m = 1$ to get the most updates possible for the same amount of compute?

Bayes Classifier / kNN:

- The Bayes classifier assigns x to the class with the largest true conditional probability $P(Y = c \mid X = x)$.
- It minimizes the expected classification error. Its error can be positive when classes overlap.
- kNN estimates these probabilities using the proportion of each class among the K nearest training observations.
- Smaller K gives a more flexible boundary, typically with lower bias and higher variance. Select K using validation or CV.
- Predictor scales matter because kNN uses distances.

Frame Title

Practice Question (5 min):

1. At a point x, the three class probabilities are 0.40, 0.35, and 0.25. What does the Bayes classifier predict, and what is its conditional error probability?
2. A 1-NN classifier has zero training error (no distance ties). Does this imply zero test error? Explain.

Logistic Regression: Binary and Multiclass

- Binary logistic regression models the log-odds linearly:

$$\log \frac{p(x)}{1-p(x)} = \beta_0 + x^\top \beta, \qquad p(x) = \frac{e^{\beta_0 + x^\top \beta}}{1 + e^{\beta_0 + x^\top \beta}}.$$

- Holding other predictors fixed, a one-unit increase in x_j multiplies the odds by e^{β_j}.
- For C classes, use class C as the reference:

$$\log \frac{P(Y = c \mid x)}{P(Y = C \mid x)} = \beta_{0c} + x^\top \beta_c, \quad c = 1, \ldots, C-1.$$

- Predict the class with the largest estimated probability. In binary classification, the threshold controls the trade-off between false positives and false negatives.

Logistic Regression: Binary and Multiclass

Practice Question (5 min):

1. Does $\beta_j = 0.7$ mean that probability increases by 0.7? Explain.
2. Why can't we fit separate binary models for each class and assume their probabilities sum to one?

LDA / QDA:

- Both assume multivariate normal predictors within each class: $X \mid Y = c \sim N_p(\mu_c, \Sigma_c)$.
- Each class has its own mean vector and prior. Bayes' rule gives

$$P(Y = c \mid X = x) = \frac{\pi_c f_c(x)}{\sum_{\ell=1}^{C} \pi_\ell f_\ell(x)}.$$

- LDA assumes a common covariance matrix $\Sigma_c = \Sigma$, giving linear decision boundaries.
- QDA estimates a separate covariance matrix for each class, giving quadratic decision boundaries.
- QDA is more flexible but estimates more parameters. LDA may perform better with smaller training samples.

LDA / QDA

Practice Question (5 min):

1. Two classes have very different covariance matrices. Why might QDA improve on LDA? Why is this not guaranteed?
2. Does equal covariance imply equal class means or priors?